

Workshop: Constrained Optimisation

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

1. Given the function:

$$z = x^2 + 2xy + 4y + 10$$

With constraint:

$$x + y = 100$$

Optimise the function z subject to the constraint:

- a. Write down the Lagrangian function

$$L(x, y, \lambda) = x^2 + 2xy + 4y + 10 + \lambda(x + y - 100)$$

- b. Write down the three first order conditions

$$L_x = 2x + 2y + \lambda = 0$$

$$L_y = 2x + 4 + \lambda = 0$$

$$L_\lambda = x + y - 100 = 0$$

- c. Solve the first order conditions to find the optimal x , y , and the value of z .

$$L_x - L_y: 2y - 4 = 0 \Rightarrow y = 2$$

Sub into constraint:

$$x + 2 = 100$$

$$x = 98$$

Sub into z

$$z = 98^2 + 2 \times 98 \times 2 + 4 \times 2 + 10 = 9818$$

2. Given the function:

$$z = 12x + 4y$$

With constraint:

$$x^{0.3}y^{0.7} = 1000$$

Optimise the function z subject to the constraint:

- a. Write down the Lagrangian function

$$L = 12x + 4y + \lambda(x^{0.3}y^{0.7} - 1000)$$

- b. Write down the three first order conditions

$$\begin{aligned}
L_x &= 12 + 0.3\lambda x^{-0.7}y^{0.7} = 0 \\
L_y &= 4 + 0.7\lambda x^{0.3}y^{-0.3} = 0 \\
L_\lambda &= x^{0.3}y^{0.7} - 1000 = 0
\end{aligned}$$

- c. Solve the first order conditions to find the optimal x , y , and the value of z .

Rearrange first two:

$$\begin{aligned}
0.3\lambda x^{-0.7}y^{0.7} &= -12 \\
0.7\lambda x^{0.3}y^{-0.3} &= -4
\end{aligned}$$

Divide one of above by the other:

$$\begin{aligned}
\frac{3}{7}x^{-1}y^1 &= \frac{12}{4} \\
\Rightarrow y &= 7x
\end{aligned}$$

Sub this into constraint:

$$\begin{aligned}
x^{0.3}(7x)^{0.7} &= 1000 \\
7^{0.7}x &= 1000 \\
x &= \frac{1000}{7^{0.7}} = 256.1 \\
y &= 7x = 1792.8 \\
z &= 12x + 4y = 10244
\end{aligned}$$

3. Given the function:

$$z = x^{0.6}y^{0.4}$$

With constraint:

$$5x + 8y = 120$$

Optimise the function z subject to the constraint

$$\begin{aligned}
L &= x^{0.6}y^{0.4} + \lambda(5x + 8y - 120) \\
L_x &= 0.6x^{-0.4}y^{0.4} + 5\lambda = 0 \\
L_y &= 0.4x^{0.6}y^{-0.6} + 8\lambda = 0 \\
L_\lambda &= 5x + 8y - 120 = 0
\end{aligned}$$

$$\begin{aligned}
5\lambda &= -0.6x^{-0.4}y^{0.4} \\
8\lambda &= -0.4x^{0.6}y^{-0.6}
\end{aligned}$$

$$\frac{5}{8} = \frac{3}{2}x^{-1}y^1$$

$$y = \frac{5}{12}x$$

$$5x + 8\left(\frac{5}{12}x\right) = 120$$

$$5x + \frac{10}{3}x = 120$$

$$\frac{25}{3}x = 120$$

$$x = 14.4$$

$$y = 6$$